

Description of Proposed Swarm Task

1. System Overview

This project proposes a **Drone Swarm for Flood Search & Rescue System** aimed at supporting emergency response operations during flood disasters in Malaysia.

The system leverages **multi-agent swarm intelligence**, where multiple autonomous drones cooperate to perform large-scale search and rescue tasks. Compared to a single-drone system, the swarm approach enables parallel exploration of large flooded areas, faster victim detection, and improved operational reliability even if individual drones fail.

The system developed will include two complementary computational paradigms:

- **Agent-Based Modelling (ABM):** Each drone will be an autonomous agent with individual decision making logic, sensor models, battery constraints and role-specific behaviour. Overall swarm coordination will be from local agent interactions rather than from centralized command.
- **Discrete Event Simulation (DES):** This simulation includes the communication infrastructure and event-driven state transitions such as victim detection or low-battery call back to capture realistic timing and model as an event queue.

The system is designed as a simulation-based prototype, with potential scalability toward real-world deployment using unmanned aerial vehicle (UAV) platforms.

2. Main Objectives

The swarm system aims to:

- Improve search efficiency and overall area coverage
 - Minimize rescue response time
 - Enable real-time victim detection and localization
 - Reduce redundant search paths through intelligent coordination
 - Provide situational awareness to rescue teams
 - Ensure safe navigation in dynamic and obstacle-filled environments
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3. System Architecture

1. Multi-Agent Structure

The system follows a hybrid coordination model:

- A leader drone provides global coordination
- Follower drones perform decentralized decision-making

2. Communication Layer

Drones exchange information through shared data structures within the simulation, simulating wireless communication such as broadcasting position updates and detected targets.

3. Data Flow

Sensor input is processed by each drone to make decisions, which are then translated into movement actions. Feedback from the environment continuously updates the system.

4. Drone Roles

Leader Drone

- Coordinates overall swarm movement
- Assigns and adjusts search areas dynamically
- Monitors swarm status and performance
- Reallocates drones when coverage gaps are detected

Follower Drones

- Explore assigned areas autonomously
- Perform obstacle avoidance and navigation
- Detect and report victims
- Share environmental data with the swarm
- Continue operation independently if communication is disrupted

Drone Agent State

State Variable	Description
Position	Contains (x,y) coordinates on the map grid.
Local Map	Have basic/partial knowledge of the terrain based on sensors and broadcast received.
Battery Level	Batter power modelled as a depletable resource; depletion rate varies by speed and payload.
Communication Range	Maximum radius within which the drone can transmit and receive messages.
Neighbour List	Dynamically maintained list of drones within communication range.

5. Search and Rescue Operations

Victim Detection

Drones scan flooded areas for stranded victims, represented as visual markers within the simulation. Upon detection, the victim's location is automatically recorded and shared with the swarm.

Area Coverage Optimization

The search area is divided among drones to minimize overlap and ensure efficient coverage. This allows faster exploration of large environments.

Obstacle Avoidance

Drones avoid both static obstacles such as buildings and debris, and dynamic obstacles such as other drones, ensuring safe operation.

Dynamic Route Adjustment

Drones continuously adjust their paths based on environmental conditions and detected obstacles, improving navigation efficiency during the mission.

6. Swarm Intelligence Features

The system applies swarm intelligence principles inspired by natural collective behaviors.

Core Algorithms

Flocking (Boids Model)

- Separation: avoids collisions
- Alignment: matches movement direction
- Cohesion: maintains group structure

Area Partitioning

- Automatically divides the search space
- Reduces redundant coverage

Cooperative Mapping

- Drones share environmental information

- Builds a collective understanding of the search area

Decentralized Decision-Making

- Each drone reacts to local conditions
 - Enhances scalability and adaptability
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7. Prototype Development

Technologies Used

- Python
- Pygame or Matplotlib for visualization
- NumPy for efficient computation

Additional simulation tools may be integrated to enhance realism and system complexity.

8. Simulation Workflow

1. Generate a flood environment with obstacles and victims
 2. Deploy drones with initial positions and roles
 3. Divide the search area automatically
 4. Begin autonomous exploration
 5. Apply flocking and collision avoidance behaviors
 6. Detect and record victim locations
 7. Update shared information across the swarm
 8. Continue searching until the entire area is covered
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9. Performance Metrics

The system performance can be evaluated using:

Metric	Definition & Significance
Coverage Rate	Percentage of non-obstacle terrain cells explored over simulation time.
Time-to-Complete	Elapsed time from mission start to all victims confirmed located and reached by Rescue drones.
Degree of overlap in search paths	Fraction of explored cells visited by more than one drone. Lower is better; indicates coordination efficiency.

Total distance traveled by drones	The total distance and radius which was covered by the drone swarms.
Collision / Near-Miss Count	Number of inter-drone proximity violations below the safety threshold per mission.

10. Expected Outcome

Upon completion of the project, the project is expected to deliver the following outcomes:

- A fully operational python simulation of an autonomous drone swarm SAR system, integrating ABM and DES in a complete framework.
- Improve efficiency in flood search operations
- Reduce response time for locating victims
- Demonstrate effective coordination among multiple drones
- Provide a scalable and fault-tolerant search strategy

The project also highlights the potential of AI-driven drone swarms in enhancing future disaster management and emergency response systems.

11. Future Work

The simulation prototype to be developed is made to be extensible for the future and can be improved for natural progressions beyond the current project scope:

- **Integration with real UAV hardware and GPS systems:** Ported for deployment on physical UAV platforms with onboard GPS or RF communication
- **Implementation of computer vision for realistic victim detection:** Replace the simulation sensor model with object detection to process real imagery for victim identification
- **Incorporation of machine learning for adaptive path planning:** Incorporate a reinforcement learning agent into the drones to allow real-time path optimization.
- **Energy-aware routing based on battery levels:** Optimize energy use in mission planning by accounting for different wind speeds, payload weight etc.
- **Simulation of communication delays and signal loss:** Replace simple DES latency parameters with real network simulator for RF modelling.
- **Expansion to multi-swarm coordination:** Simulate multiple independent swarms operating together over different sectors.
- **Extension to three-dimensional environments with altitude control:** Extend the grid from 2D to 3D to support altitude navigation, obstacle avoidance etc.